

SKILL-week4

Task-1: Tokenization

```
root@Sahasra:~# cd ~
mkdir -p skill_week4
cd skill_week4
root@Sahasra:~# pwd
/home/sahasra/skill_week4
root@Sahasra:~# touch myshell.c
root@Sahasra:~# nano myshell.c
root@Sahasra:~# gcc myshell.c -o myshell
root@Sahasra:~# ./myshell
myshell$ echo Hello
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello
Parent Process PID: 542
Created Child PID: 543
Child Process PID: 543
Hello
Child exited with status: 0
myshell$ echo Hello World
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello
Argument 3: World
Parent Process PID: 542
Created Child PID: 544
Child Process PID: 544
Hello World
Child exited with status: 0
myshell$ pwd
--- Parsed Arguments ---
Argument 1: pwd
Parent Process PID: 542
Created Child PID: 545
Child Process PID: 545
/home/sahasra/skill_week4
Child exited with status: 0
myshell$ ls
--- Parsed Arguments ---
Argument 1: ls
Parent Process PID: 542
Created Child PID: 548
Child Process PID: 548
myshell myshell.c myshell.c.save
Child exited with status: 0
```

Task-2: Single Quotes

```

root@Sahasra:~# touch myshell.c
root@Sahasra:~# nano myshell.c
root@Sahasra:~# gcc myshell.c -o myshell
root@Sahasra:~# ./myshell
myshell$ echo 'Hello World'
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello World
Parent Process PID: 574
Created Child PID: 575
Child Process PID: 575
Hello World
Child exited with status: 0
myshell$ echo '$USER'
--- Parsed Arguments ---
Argument 1: echo
Argument 2: $USER
Parent Process PID: 574
Created Child PID: 578
Child Process PID: 578
$USER
Child exited with status: 0
myshell$ echo 'Hello World
Syntax Error: Unmatched single quote.

```

Task-3: Double Quotes

```

root@Sahasra:~# touch myshell.c
root@Sahasra:~# nano myshell.c
root@Sahasra:~# gcc myshell.c -o myshell
root@Sahasra:~# ./myshell
myshell$ echo "Hello World"
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello World
Parent Process PID: 604
Created Child PID: 606
Child Process PID: 606
Hello World
Child exited with status: 0
myshell$ echo "Hello $USER"
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello sahasra
Parent Process PID: 604
Created Child PID: 607
Child Process PID: 607
Hello sahasra
Child exited with status: 0
myshell$ echo "Hello World
Syntax Error: Unmatched double quote.

```

Task-4: Escaped Space

```

root@Sahasra:~# touch myshell.c
root@Sahasra:~# nano myshell.c
root@Sahasra:~# gcc myshell.c -o myshell
root@Sahasra:~# ./myshell
myshell$ echo Hello\ World
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello World
Parent Process PID: 626
Created Child PID: 627
Child Process PID: 627
Hello World
Child exited with status: 0
myshell$ echo Hello\@World
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello@World
Parent Process PID: 626
Created Child PID: 628
Child Process PID: 628
Hello@World
Child exited with status: 0
myshell$ echo \$USER
--- Parsed Arguments ---
Argument 1: echo
Argument 2: $USER
Parent Process PID: 626
Created Child PID: 629
Child Process PID: 629
$USER
Child exited with status: 0

```

Task-5: Child Process

```

root@Sahasra:~# ./myshell
myshell$ ls
--- Parsed Arguments ---
Argument 1: ls
Parent Process PID: 665
Created Child PID: 666
Child Process PID: 666
myshell myshell.c myshell.c.save
Child exited with status: 0
myshell$ pwd
--- Parsed Arguments ---
Argument 1: pwd
Parent Process PID: 665
Created Child PID: 668
Child Process PID: 668
/home/sahasra/skill_week4
Child exited with status: 0
myshell$ whoami
--- Parsed Arguments ---
Argument 1: whoami
Parent Process PID: 665
Created Child PID: 669
Child Process PID: 669
sahasra
Child exited with status: 0
myshell$ date
--- Parsed Arguments ---
Argument 1: date
Parent Process PID: 665
Created Child PID: 670
Child Process PID: 670
Mon Aug 17 05:16:20 UTC 2026
Child exited with status: 0
myshell$ exit
Shell exiting...

```

```
root@Sahasra:~# ./myshell
myshell$ ls
--- Parsed Arguments ---
Argument 1: ls
Parent Process PID: 665
Created Child PID: 666
Child Process PID: 666
myshell myshell.c myshell.c.save
Child exited with status: 0
myshell$ pwd
--- Parsed Arguments ---
Argument 1: pwd
Parent Process PID: 665
Created Child PID: 668
Child Process PID: 668
/home/sahasra/skill_week4
Child exited with status: 0
myshell$ whoami
--- Parsed Arguments ---
Argument 1: whoami
Parent Process PID: 665
Created Child PID: 669
Child Process PID: 669
sahasra
Child exited with status: 0
myshell$ date
--- Parsed Arguments ---
Argument 1: date
Parent Process PID: 665
Created Child PID: 670
Child Process PID: 670
Mon Aug 17 05:16:20 UTC 2026
Child exited with status: 0
myshell$ exit
Shell exiting...
```